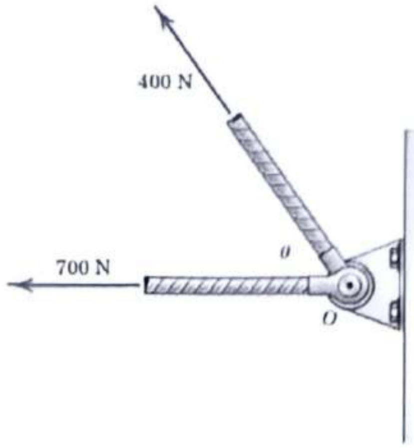
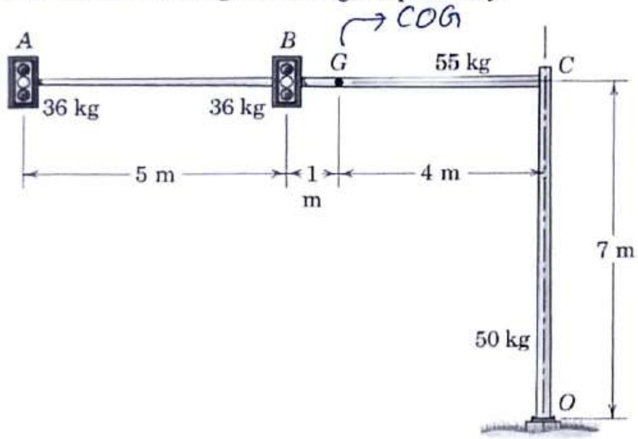
Indian Institute of Technology Jodhpur
NH 65, Surpura Bypass Rd, Karwar, Rajasthan 342037

Course: Engineering Mechanics | **Code: MEL1010**
Date: 18/02/2026 | **Maximum Marks: 20**

Question 1: At what angle θ must the 400-N force be applied in order that the resultant R of the two forces have a magnitude of 1000 N? [1]

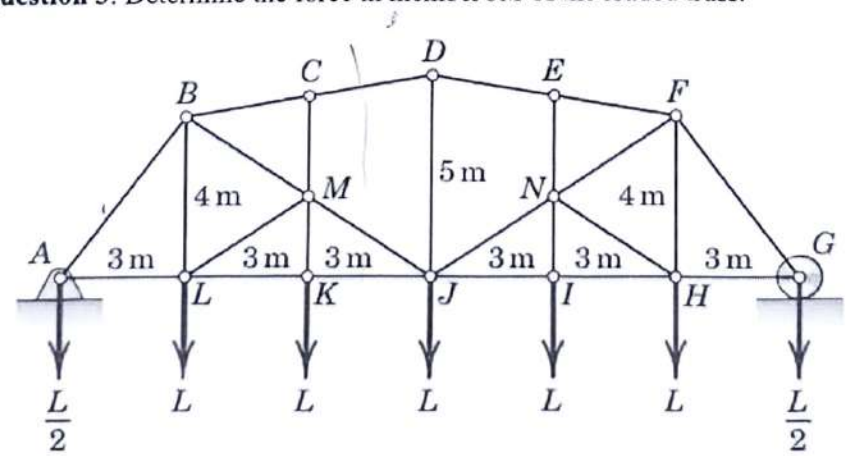


Question 2: Calculate the force and moment reactions at the bolted base O of the overhead traffic-signal assembly. Each traffic signal has a mass of 36 kg, while the masses of members OC and AC are 50 kg and 55 kg, respectively.



[3]

Question 3: Determine the force in member *JM* of the loaded truss. [4]



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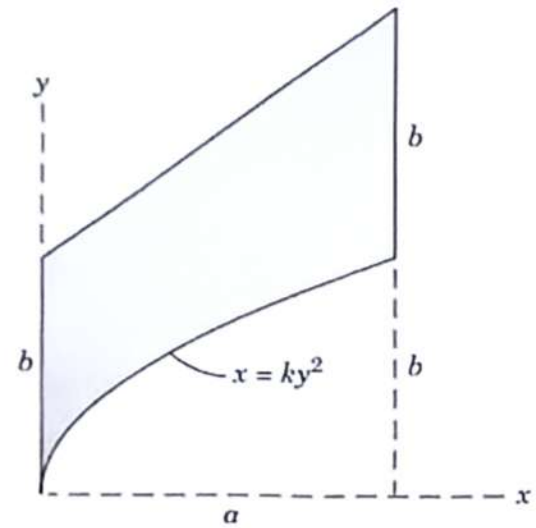
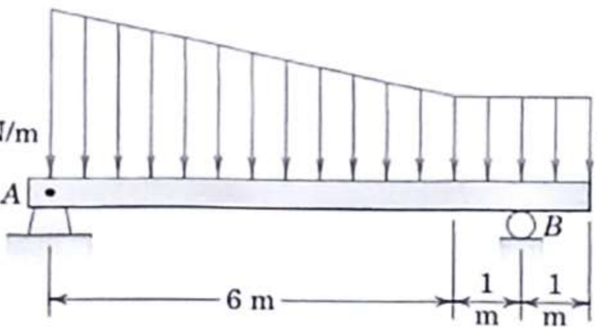
Question 4: Determine the y-coordinate of the centroid of the shaded area. [4]

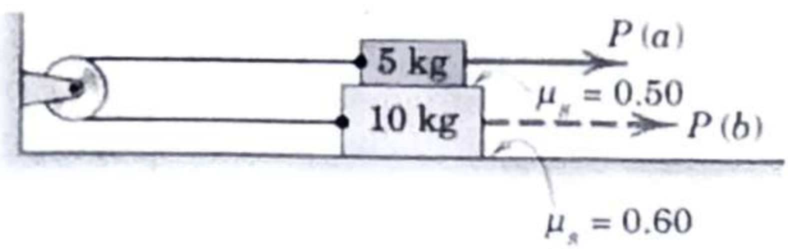


Question 5: Determine the reactions at the supports of the beam, which is loaded as shown.



[4]

Question 6: The system of two blocks, cable, and fixed pulley is initially at rest. Determine the horizontal force P necessary to cause motion when (a) P is applied to the 5-kg block and (b) P is applied to the 10-kg block. Determine the corresponding tension T in the cable for each case.



[4]

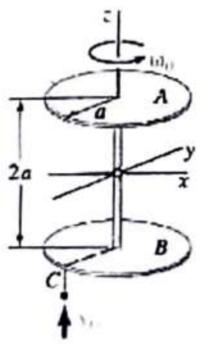
Indian Institute of Technology Jodhpur

Department of Mechanical Engineering, Nagour Road, Karwar, Jodhpur 342037, Rajasthan, INDIA

Course: Engineering Mechanics Major 22nd April 2026 Code: MEL 1010 40 Marks

Do not ask doubts during the major exam. Make suitable assumptions. Assume missing data. Final answer must be computed with correct numbers and units. Otherwise no marks will be awarded.

1. Inspired by the song "Haan Hai Koi To Wajah Jo Jeena Ka Maza Yun Aane Laga…Yeh Ishq Haaye" a couple decided to drive through the high altitude Rohtang pass. Assume the following data for the vehicle used in the song when climbing the hill: The center of mass of the vehicle is at a distance 1.1m from the front wheels and 1.275 m from the rear wheels. It has 2.375m wheelbase, width 1540 mm, 79hp engine and a gross weight of 1500 kg. Height of center of mass of the vehicle above flat ground is about 1.0m; Assume the road grade/incline is 5.71^0 degrees. Take Friction coefficient=0.3; Compute maximum acceleration of the four wheeler (i) with an all-wheel drive (ii) rear-wheel drive and (iii) a front wheel drive? [9]
2. A cyclone builds up in the Bay of Bengal. The center of the hurricane is 100 km from the eastern coast of India and moving at the speed of 50 km/hr to the west along the latitude 20^0N. Taking only coriolis force into account we wish to determine whether the hurricane would hit the coast. What is the trajectory of the eye of the hurricane? What is the equation of the trajectory? In which sense do the winds around the hurricane eye move as seen from a satellite? Doe the hurricane eye hit the coast? [6]
3. Students are testing a drone for quick commerce applications as the working conditions of delivery personnel do not allow under the '10-minute delivery' for road based delivery. The position data of the drone can be approximated using the expressions: horizontal $x(t) = -0.0000225t^4 + 0.003t^3 + 0.01t^2$ and vertical $y(t) = 300[1-\cos(\pi t/40)]$, where x and y are in meters and t in seconds. Knowing that the take-off and landing altitudes are the same, determine (a) Duration of flight (b) its maximum altitude and (c) the horizontal range travelled during the flight. Is this an energy efficient mode of delivery? [4]
4. A simple gyroscope with a symmetric rotor/disk is filmed at 600 frames/sec. The disk rotates once every 10 frames. It precesses once in 1400 frames. The mass of the disk is 108 grams, and its diameter is 56 mm. Find the spin speed, precession speed and the torque acting on the rotor/disk. Is this result exact? What are the sources of error? [4]
5. James webb space telescope inertia matrix components (in Kg m^2) are given as I_{xx}= 67946; I_{xy}= −83; I_{xz} =11129; I_{yy}=90061; I_{yz}=103; I_{zz}=45821; Explain whether this a valid inertia matrix? Compute the principle moments of inertia. Determine the axis about which it is unstable. [6]
6. A space satellite with mass m can be modelled as two thin disks of equal mass. The disks have a radius of a= 800 mm and are rigidly connected by a light rod with a length of 2a. Initially, the satellite is spinning freely about its axis of symmetry at the rate ω_0=60 rpm. A meteorite with a mass of m_0 =m/1000 is traveling with a velocity v_0 of 2000 m/s relative to the satellite, strikes the satellite, and becomes embedded at C. Determine (a) the angular velocity of the satellite immediately after impact, (b) the precession axis of the ensuing motion; [6]



7. The jet aircraft has a mass of 4.6 Mg and a drag (air resistance) of 32 kN at a speed of 1000 km/h at a particular altitude. The aircraft consumes air at the rate of 106 kg/s through its intake scoop and uses fuel at the rate of 4 kg/s. If the exhaust has a rearward velocity of 680 m/s relative to the exhaust nozzle, determine the maximum angle of elevation α at which the jet can fly with a constant speed of 1000 km/h at the particular altitude in question. [5]

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Course: Engineering Mechanics
Date: 22/03/2026

Code: MEL1010
Quiz 2 [20 marks]

4. In early-mid 2024, 12 Starlink satellites were lost in the low Earth orbit (at lower altitudes 300 – 400 km). The cause for this loss is believed to be a deviation from central force motion assumptions. Write the central force motion dynamic equations using Newton's laws in polar coordinates and explain the possible deviations which caused loss of satellites. [5]
5. The goal keeper kicks the soccer ball into a heavy rain such that air resists the motion in a manner proportional to the velocity squared. If the initial velocity of the ball is 30 m/s and the ball leaves the keeper's foot at a 45 degrees angle, determine the distance the ball travels for two cases.
   The acceleration vector is given by $-g\,j - cv\,|v|$ where (i) c=0 when there is no rain and (ii) c =0.005 (m^{-1}) during rain. [10]
6. During a "lock up skid", a car has a constant negative acceleration. If there is a 1.75seconds reaction time of the driver before braking and deceleration is $6.86 m/s^2$, determine the distance to stop if the initial speed is (a) 100km/hr; (b) 50km/hr. Name an automotive braking invention that may prevent such skids. [5]